

CO3-AT1

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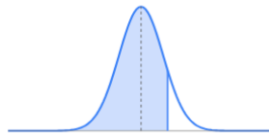
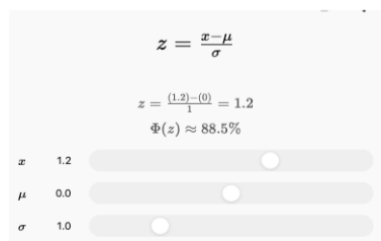
1. Detecting an Unusual Customer

A shopping website finds that the average amount spent by customers is ₹2,000, with a

standard deviation of ₹400.

A particular customer spends ₹3,200.

1. Calculate the customer's z-score.
2. Is the customer's spending unusually high compared with the average?
3. What would a negative z-score mean in this context?
4. If another customer has a z-score of -1.5 , what does it tell you?



Sol:

Given:

- Average spending (Mean, μ) = ₹2,000
- Standard deviation (σ) = ₹400
- Customer spending (x) = ₹3,200

1. Calculate the customer's z-score

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Substitute the values:

$$z = \frac{3200 - 2000}{400} = \frac{1200}{400} = 3$$

Answer: $z = 3$

2. Is the customer's spending unusually high?

A **z-score of 3** means the customer's spending is **3 standard deviations above the average**.

Normally:

- $z = 0 \rightarrow$ Average
- $z = \pm 1 \rightarrow$ Normal
- $z = \pm 2 \rightarrow$ High/Low
- **$z = \pm 3$ or more $\rightarrow$ Very unusual**

Answer: Yes. Spending ₹3,200 is **unusually high** compared to the average customer.

3. What would a negative z-score mean?

A **negative z-score** means the customer spent **less than the average**.

Example:

- $z = -1 \rightarrow$ Spending is 1 standard deviation below the mean.
- $z = -2 \rightarrow$ Spending is much lower than average.

Answer: A negative z-score indicates **below-average spending**.

4. If another customer has a z-score of -1.5 , what does it tell you?

A z-score of **-1.5** means the customer spent **1.5 standard deviations below the average**.

It shows:

- Spending is lower than average.
- It is not extremely unusual but noticeably lower.

Answer: The customer spends **less than most customers**.

2. Delivery Time Analysis

A delivery company records delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the mean and median.
2. Find Q1 and Q3.
3. Calculate the IQR.
4. Check for outliers using the $1.5 \times \text{IQR}$ rule.
5. Explain whether the mean or median gives a better description of the typical delivery time.

Sol:

Delivery times (minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Mean and Median

Mean

$$\text{Mean} = \frac{25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60}{10} = \frac{326}{10} = 32.6$$

Mean = 32.6 minutes

Median

There are 10 values (even number).

Median is the average of the 5th and 6th values.

$$5\text{th} = 30$$

$$6\text{th} = 30$$

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Median = 30 minutes

2. Find Q1 and Q3

Lower half:

$$25, 27, 28, 29, 30$$

Middle value = **28**

Q1 = 28

Upper half:

$$30, 31, 32, 34, 60$$

Middle value = **32**

Q3 = 32

3. Calculate IQR

Formula:

$$IQR = Q3 - Q1$$

$$IQR = 32 - 28 = 4$$

IQR = 4

4. Check for Outliers

Lower limit:

$$Q1 - 1.5(IQR) = 28 - 1.5(4) = 22$$

Upper limit:

$$Q3 + 1.5(IQR) = 32 + 6 = 38$$

Values greater than **38** are outliers.

Only **60** is above 38.

Answer: 60 minutes is an outlier.

5. Which is better: Mean or Median?

Because **60** is an outlier, it increases the mean.

Mean = 32.6

Median = 30

The **median** is not affected much by the outlier.

Answer: The median better represents the typical delivery time.

3. Two Cities's Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

1. Calculate the mean temperature for each city.
2. Calculate the variance for each city.
3. Which city has greater temperature variability?
4. Why can two datasets have the same mean but different standard deviations?
5. Which city has more predictable temperatures?

Sol:

City A

28, 29, 30, 30, 31, 30, 32

City B

20, 25, 30, 35, 30, 25, 45

1. Mean Temperature

City A

$$\frac{28 + 29 + 30 + 30 + 31 + 30 + 32}{7} = \frac{210}{7} = 30^{\circ}\text{C}$$

City B

$$\frac{20 + 25 + 30 + 35 + 30 + 25 + 45}{7} = \frac{210}{7} = 30^{\circ}\text{C}$$

Answer:

- City A = **30°C**
- City B = **30°C**

2. Variance

Variance measures how much the temperatures spread around the mean.

City A

Squared deviations:

$$(-2)^2 + (-1)^2 + 0^2 + 0^2 + 1^2 + 0^2 + 2^2 = 4 + 1 + 0 + 0 + 1 + 0 + 4 = 10$$

Population variance:

$$\frac{10}{7} \approx 1.43$$

City B

Squared deviations:

$$\begin{aligned} &(-10)^2 + (-5)^2 + 0^2 + 5^2 + 0^2 + (-5)^2 + 15^2 \\ &100 + 25 + 0 + 25 + 0 + 25 + 225 = 400 \end{aligned}$$

Population variance:

$$\frac{400}{7} \approx 57.14$$

3. Which city has greater temperature variability?

Compare variances:

- City A = **1.43**
- City B = **57.14**

Answer: City B has much greater temperature variability.

4. Why can two datasets have the same mean but different standard deviations?

The **mean** only tells us the average value.

The **standard deviation (or variance)** tells us how spread out the values are.

Although both cities have the same average temperature (**30°C**):

- City A temperatures stay close to 30°C.
- City B temperatures vary widely from 20°C to 45°C.

Answer: Two datasets can have the same average but different spreads, resulting in different standard deviations.

5. Which city has more predictable temperatures?

A smaller variance means temperatures change less from day to day.

- City A → Low variance → More consistent.
- City B → High variance → Less consistent.

Answer: City A has more predictable temperatures because its temperatures are more consistent.

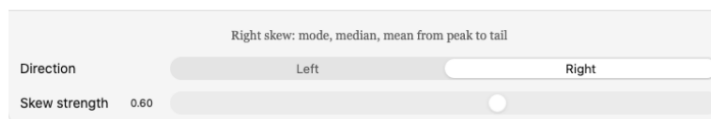
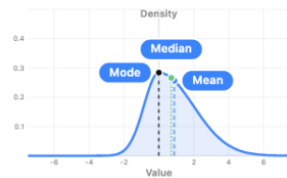
4. Data Center CPU Usage

A data scientist records CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate mean, median, and mode.
2. Calculate the range.

3. Find Q1, Q3 and IQR.
4. Identify possible outliers.
5. Determine the direction of skewness.



Given Data

CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate Mean, Median, and Mode

Mean

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Sum:

$$40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90 = 513$$

$$\text{Mean} = \frac{513}{10} = 51.3$$

Mean = 51.3%

Median

There are **10 values** (even number).

Median = Average of the 5th and 6th values.

5th value = **46**

6th value = **48**

$$\text{Median} = \frac{46 + 48}{2} = 47$$

Median = 47%

Mode

The value **45** appears **twice**, while all others appear once.

Mode = 45%

2. Calculate the Range

Formula:

$$\begin{aligned}\text{Range} &= \text{Maximum} - \text{Minimum} \\ 90 - 40 &= 50\end{aligned}$$

Range = 50%

3. Find Q1, Q3 and IQR

Lower Half

40, 42, 45, 45, 46

Middle value = **45**

Q1 = 45

Upper Half

48, 50, 52, 55, 90

Middle value = **52**

Q3 = 52

IQR

$$\begin{aligned}IQR &= Q3 - Q1 \\ 52 - 45 &= 7\end{aligned}$$

IQR = 7

4. Identify Possible Outliers

Lower Limit

$$Q1 - 1.5(IQR)$$
$$45 - 1.5(7) = 45 - 10.5 = 34.5$$

Upper Limit

$$Q3 + 1.5(IQR)$$
$$52 + 10.5 = 62.5$$

Any value greater than **62.5** is an outlier.

Only **90** is greater than 62.5.

Answer: 90% is an outlier.

5. Determine the Direction of Skewness

Since **90** is much larger than the other values, it pulls the mean toward the right.

Notice:

- Mode = **45**
- Median = **47**
- Mean = **51.3**

Since

$$\text{Mean} > \text{Median} > \text{Mode}$$

the distribution is **positively (right) skewed**.

Answer: The dataset is right-skewed (positively skewed).

5. Comparing Two Datasets

Two data analysts receive the following datasets:

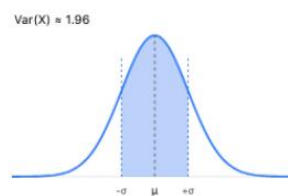
Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

1. Calculate the mean of both datasets.
2. Calculate the variance of both.
3. Calculate the standard deviation of both.
4. Which dataset has greater spread?
5. If both datasets represent model errors, which model appears more consistent?
6. Explain why mean alone is not enough to compare datasets.

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$
$$\text{Var}(X) = (1.4)^2 = 1.96$$

σ 1.4



Sol:

Given

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

1. Calculate the Mean

Dataset A

$$\frac{10 + 20 + 30 + 40 + 50}{5} = \frac{150}{5} = 30$$

Mean = 30

Dataset B

$$\frac{28 + 29 + 30 + 31 + 32}{5} = \frac{150}{5} = 30$$

Mean = 30

2. Calculate the Variance

Dataset A

Mean = 30

Squared deviations:

$$(10 - 30)^2 = 400$$

$$(20 - 30)^2 = 100$$

$$(30 - 30)^2 = 0$$

$$(40 - 30)^2 = 100$$

$$(50 - 30)^2 = 400$$

Total = **1000**

Population Variance:

$$\frac{1000}{5} = 200$$

Variance = 200

Dataset B

Squared deviations:

$$(28 - 30)^2 = 4$$

$$(29 - 30)^2 = 1$$

$$(30 - 30)^2 = 0$$

$$(31 - 30)^2 = 1$$

$$(32 - 30)^2 = 4$$

Total = **10**

Population Variance:

$$\frac{10}{5} = 2$$

Variance = 2

3. Calculate the Standard Deviation

Dataset A

$$\sqrt{200} \approx 14.14$$

Standard Deviation ≈ 14.14

Dataset B

$$\sqrt{2} \approx 1.41$$

Standard Deviation ≈ 1.41

4. Which Dataset Has Greater Spread?

Compare standard deviations:

- Dataset A = **14.14**
- Dataset B = **1.41**

Answer: Dataset A has a much greater spread because its values are farther from the mean.

5. If Both Datasets Represent Model Errors, Which Model Is More Consistent?

A smaller standard deviation means the values are closer to the mean and therefore more consistent.

- Dataset A $\rightarrow$ Large variation.
- Dataset B $\rightarrow$ Small variation.

Answer: Dataset B is more consistent because its errors vary very little.

6. Why Is Mean Alone Not Enough to Compare Datasets?

Both datasets have the same mean (**30**), but their values are distributed very differently:

- **Dataset A** ranges widely from **10 to 50**, indicating high variability.
- **Dataset B** stays close to **30**, indicating low variability.

Therefore, the **mean alone does not show how spread out the data is**.
Measures such as **variance** and **standard deviation** are needed to understand consistency and variability.